Catalyst. *by Mrs. Rana*

GRADE 10 · MATH · TERM 1

REVEAL MATH · GENERAL

Revision Booklet

Every lesson of the term on one page per part, plus a Last Look for the night before the exam.

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Scan me

The course site: every lesson with the mind map, flashcards, traps and new practice questions every time.

LESS EFFORT. FULL MARK.

Line Them Up

Adding and Subtracting Polynomials

SCAN FOR PRACTICE

MUST KNOW

1 Is it a polynomial?

1. Look at every exponent on a variable. It must be 0, 1, 2, 3...
كل أس لازم يكون عدد كلي.
2. A negative exponent means the variable is in the denominator → **not** a polynomial.
الأس السالب = متغير في المقام ← ليست كثيرة حدود.
3. A number alone is a polynomial: **−11.25** is a monomial of degree 0.
العدد وحده كثيرة حدود.

Is $2x^{-2} + 3xy$ a polynomial?

$2x^{-2} = \frac{2}{x^2}$ — negative exponent → denominator

Answer: No

2 Count the terms

1. Combine like terms first.
اجمع الحدود المتشابهة أولًا.
2. Count what is left: 1 monomial, 2 binomial, 3 trinomial.
١ أحادية، ٢ ثنائية، ٣ ثلاثية.

Classify $9x^3 - 8x + 5x - 27$.

$-8x + 5x = -3x$ — combine like terms

$9x^3 - 3x - 27$ — 3 terms left

Answer: trinomial (not 4 terms)

3 Find the degree

1. For each term, add the exponents of all its variables.
لكل حد: اجمع أسس متغيراته.
2. The degree of the polynomial is the biggest of these.
درجة كثيرة الحدود = أكبر رقم.

Find the degree of $6a^2b^2 - 2ab + 5ab^2$.

$6a^2b^2 \rightarrow 2 + 2 = 4$ — term 1

$-2ab \rightarrow 1 + 1 = 2 \cdot 5ab^2 \rightarrow 1 + 2 = 3$ — terms 2 and 3

Answer: degree 4 (quartic)

4 Standard form & leading coefficient

1. Write the degree above each term.
اكتب درجة كل حد.
2. Rewrite from biggest degree to smallest. The sign moves with its term.
رتّب من الأكبر للأصغر، والإشارة تمشي مع حدها.
3. The number in front of the first term is the leading coefficient.
معامل الحد الأول = المعامل الرئيسي.

Write $4x + 12 + 2x^3 - 3x^2$ in standard form.

degrees: $4x \rightarrow 1 \cdot 12 \rightarrow 0 \cdot 2x^3 \rightarrow 3 \cdot -3x^2 \rightarrow 2$ — step 1

$2x^3 - 3x^2 + 4x + 12$ — order 3, 2, 1, 0

Leading coefficient 2 · degree 3

DON'T FALL FOR IT

✗ Degree of $2a^3b^2$ is 3	✓ $3 + 2 = 5$	Add all the exponents in one term. اجمع كل أسس الحد الواحد.
✗ $9x^3 - 8x + 5x - 27$ has 4 terms	✓ 3 terms → trinomial	Combine like terms before you count. بسّط قبل ما تعدّ.
✗ LC of $-b^3 + 5b$ is 1	✓ LC = −1	The sign belongs to the term. الإشارة جزء من الحد.

UNIT 1 · LESSON 11-1 · PART B · BOOK P.631

Line Them Up

Adding and Subtracting Polynomials

SCAN FOR PRACTICE

MUST KNOW

5 Add polynomials

1. Remove the brackets (a + before a bracket changes nothing).
افتح الأقواس.
2. Group like terms: x^2 with x^2, x with x, numbers with numbers.
جمّع المتشابهات.
3. Add the coefficients. Keep the exponents. Write in standard form.
اجمع المعاملات واكتب بالصيغة القياسية.

Find $(4x^2 - 3x + 7) + (x^2 + 5x - 9)$.

$= (4x^2 + x^2) + (-3x + 5x) + (7 - 9)$ — group like terms

$= 5x^2 + 2x - 2$ — add coefficients

Answer: $5x^2 + 2x - 2$

6 Subtract polynomials

1. Change the sign of **every** term in the second bracket (the additive inverse).
غيّر إشارة كل حد في القوس الثاني.
2. Now it is an addition: group like terms.
بقت جمع: جمّع المتشابهات.
3. Combine and write in standard form.
بسّط واكتب بالصيغة القياسية.

Find $(7x - 4) - (3x - 10)$.

$= 7x - 4 - 3x + 10$ — every sign of $(3x - 10)$ changes

$= (7x - 3x) + (-4 + 10)$ — group

$= 4x + 6$ — combine

Answer: $4x + 6$

7 Vertical method

1. Write both polynomials in standard form.
اكتب الاتنين بالصيغة القياسية.
2. Line up like terms in columns. Put $0x^2$ (or $0x$) where a term is missing.
رتّب الأعمدة وحط صفر للحد الناقص.
3. For subtraction, change every sign in the bottom row, then add the columns.
في الطرح غيّر إشارات الصف التحتاني ثم اجمع.

Find $(2x^2 + 5) - (4x - x^2 + 3)$.

$2x^2 + 0x + 5$ — top, with a placeholder

$-x^2 + 4x + 3 \rightarrow$ change signs $\rightarrow +x^2 - 4x - 3$ — bottom row

$3x^2 - 4x + 2$ — add the columns

Answer: $3x^2 - 4x + 2$

8 Word problems

1. Perimeter: add all the sides.
المحيط: اجمع كل الأضلاع.
2. Profit: $P = R - C$ (revenue minus cost).
الربح = الإيراد − التكلفة.
3. "How many more" or "difference": subtract, then substitute the number if asked.
"كم يزيد": اطرح، ثم عوّض لو مطلوب.

Revenue $R = -0.02x^2 + 60x$ and cost $C = 12x + 800$.
Find the profit P.

$P = R - C = (-0.02x^2 + 60x) - (12x + 800)$ — profit formula

$= -0.02x^2 + 60x - 12x - 800$ — change signs of C

$= -0.02x^2 + 48x - 800$ — combine

Answer: $P = -0.02x^2 + 48x - 800$

DON'T FALL FOR IT

✗	✓	
$(7x - 4) - (3x - 10) = 4x - 14$	$4x + 6$	Change the sign of EVERY term in the second bracket, not only the first. غيّر إشارة كل الحدود مش أول حد بس.
$x^2 + x^2 = x^4$	$2x^2$	When you add, the exponent stays the same. في الجمع الأس ثابت.
$3x + 2y = 5xy$	$3x + 2y$ (stays)	Unlike terms cannot be combined. الحدود غير المتشابهة لا تُجمع.

LAST LOOK · LESSON 11-1 · THE NIGHT BEFORE THE EXAM

Line Them Up

Adding and Subtracting Polynomials

SCAN FOR PRACTICE

Add & Subtract Polynomials

1 Polynomial or not?

Exponents must be whole numbers 0, 1, 2... No variable in a denominator.

✓ $3x^2 + 1$ × $2x^{-2}$

الأسس أعداد كلية، ولا متغير في المقام.

2 Count the terms

1 term monomial · 2 binomial · 3 trinomial. Simplify first.

8ab − 2c → binomial

عدّ الحدود بعد التبسيط.

3 Find the degree

Term: add its exponents. Polynomial: the biggest term degree.

$2a^3b^2 \rightarrow 3 + 2 = 5$

درجة الحد = مجموع الأسس، ودرجة كثيرة الحدود = الأكبر.

4 Standard form

Biggest degree first. The first number is the leading coefficient.

$2x^3 - 3x^2 + 4x + 12$

الأكبر درجة أولًا، وأول معامل = المعامل الرئيسي.

5 Add

Remove the brackets. Combine like terms. Exponents never change.

$(4x^2 - 3x) + (x^2 + 5x) = 5x^2 + 2x$

اجمع الحدود المتشابهة، والأس ما يتغيرش.

6 Subtract

Change the sign of every term in the 2nd bracket, then add.

$-(3x - 10) = -3x + 10$

غيّر إشارة كل حدود القوس التاني ثم اجمع.

7 Vertical method

Write both in standard form. Line up like terms. Use $0x^2$ for a missing term.

$0x^2 + x + 2$

رتّب تحت بعض، وحط صفر للحد الناقص.

8 Word problems

Perimeter = add all sides. Profit = Revenue − Cost. Difference = bigger − smaller.

P = R − C

المحيط = مجموع الأضلاع، الربح = الإيراد − التكلفة.

WORDS TO KNOW

term حد	**monomial** وحيدة الحد	**polynomial** كثيرة الحدود	**binomial · trinomial** ثنائية الحد · ثلاثية الحد
degree الدرجة	**standard form** الصيغة القياسية	**leading coefficient** المعامل الرئيسي	**constant** حد ثابت
like terms حدود متشابهة	**additive inverse** المعكوس الجمعي	**sum** المجموع	**difference** الفرق
perimeter المحيط	**closed** مغلقة		

In the final exam: **Q1** · scan the QR for timed practice

UNIT 1 · LESSON 11-2 · BOOK P.639

Share It Out

Multiplying Polynomials by Monomials

SCAN FOR PRACTICE

MUST KNOW

1 Multiply a polynomial by a monomial

1. Multiply the monomial by every term in the bracket.
اضرب في كل حد.
2. Multiply the coefficients (with their signs). Add the exponents of the same variable.
اضرب المعاملات واجمع الأسس.
3. Write the answer in standard form.
اكتب بالصيغة القياسية.

Simplify $-3x(2x^2 - 5x + 4)$.

$(-3x)(2x^2) + (-3x)(-5x) + (-3x)(4)$ — distribute

$-6x^3 + 15x^2 - 12x$ — multiply, add exponents

Answer: $-6x^3 + 15x^2 - 12x$

2 Simplify expressions

1. Distribute each bracket separately. A minus before a number goes with it.
وزّع كل قوس لوحده، والسالب يمشي مع الرقم.
2. Combine like terms.
اجمع المتشابهات.
3. Write in standard form.
اكتب بالصيغة القياسية.

Simplify $2a(3a^2 - 4a) - 5(a^3 - 2)$.

$6a^3 - 8a^2 - 5a^3 + 10$ — distribute (−5 times −2 is +10)

$(6a^3 - 5a^3) - 8a^2 + 10$ — group

$a^3 - 8a^2 + 10$ — combine

Answer: $a^3 - 8a^2 + 10$

3 Solve equations

1. Distribute on both sides.
وزّع في الطرفين.
2. Combine like terms. If x^2 appears on both sides, it cancels.
بسّط، ولو $$x^2$$ في الطرفين هيتشال.
3. Move the variable to one side and divide.
اعزل المتغير.

Solve $x(x + 4) - 2 = x(x - 1) + 13$.

$x^2 + 4x - 2 = x^2 - x + 13$ — distribute

$4x - 2 = -x + 13$ — subtract x^2 from both sides

$5x = 15$ — add x and 2

$x = 3$ — divide by 5

Answer: $x = 3$

4 Area and real-life

1. Write the formula: rectangle $A = lw$, triangle $A = \frac{1}{2}bh$, trapezoid $A = \frac{1}{2}h(a + b)$.
اكتب القانون.
2. Substitute the expressions and distribute.
عوّض ووزّع.
3. If a value is given, substitute it at the end.
لو فيه قيمة عوّض في الآخر.

A rectangle has width $3x$ and length $2x + 5$. Find its area.

$A = 3x(2x + 5)$ — $A = lw$

$= 6x^2 + 15x$ — distribute

Answer: $6x^2 + 15x$

DON'T FALL FOR IT

✗	✓	
$-2x(4x^2 - 5) = -8x^3 - 10x$	$-8x^3 + 10x$	Negative × negative = positive. سالب × سالب = موجب.
$x^2 \cdot x^3 = x^6$	x^5	When you multiply, ADD the exponents. في الضرب نجمع الأسس.
$3x(2x + 1) = 6x^2 + 1$	$6x^2 + 3x$	Multiply the monomial by every term, including the number. اضرب في كل الحدود حتى الرقم.

LAST LOOK · LESSON 11-2 · THE NIGHT BEFORE THE EXAM

Share It Out

Multiplying Polynomials by Monomials

SCAN FOR PRACTICE

Monomial × Polynomial

1 Distribute

Multiply the monomial by EVERY term inside.

$3x(2x + 5) = 6x^2 + 15x$

اضرب في كل حد جوه القوس.

2 Exponents rule

Multiply coefficients, ADD exponents. Watch the signs.

$-2x^2 \cdot 4x^3 = -8x^5$

المعاملات تُضرب والأسس تُجمع.

3 Simplify expressions

Distribute each bracket, then combine like terms.

$2a(a - 1) + 3a = 2a^2 + a$

وزّع كل قوس ثم اجمع المتشابهات.

4 Solve equations

Distribute, combine, then get the variable alone.

$2p = 3(4p - 10) \rightarrow p = 3$

وزّع، بسّط، ثم اعزل المتغير.

WORDS TO KNOW

monomial وحيدة الحد	**Distributive Property** خاصية التوزيع	**product** حاصل الضرب	**like terms** حدود متشابهة
closed under multiplication مغلقة تحت الضرب			

In the final exam: **Q21** · scan the QR for timed practice

Every Term Meets

Multiplying Polynomials

SCAN FOR PRACTICE

MUST KNOW

1 Multiply two binomials (FOIL)

1. F: multiply the first terms. O: the outer terms. I: the inner terms. L: the last terms.
الأول، الخارجي، الداخلي، الأخير.
2. Keep every sign with its term.
الإشارة تمشي مع الحد.
3. Combine the two middle terms (O + I).
اجمع الحدين اللي في النص.

Find $(2a - 5)(3a + 2)$.

F: $2a \cdot 3a = 6a^2$ · O: $2a \cdot 2 = 4a$	first, outer
I: $-5 \cdot 3a = -15a$ · L: $-5 \cdot 2 = -10$	inner, last
$6a^2 + 4a - 15a - 10 = 6a^2 - 11a - 10$	combine O + I

Answer: $6a^2 - 11a - 10$

2 Vertical method

1. Write one binomial under the other.
اكتب القوسين تحت بعض.
2. Multiply the top by the last term, then by the first term (shift one column left).
اضرب في الحد الأخير ثم الأول وازّح خانة.
3. Add the columns.
اجمع الأعمدة.

Find $(x - 1)(x + 7)$ vertically.

$7(x - 1) = 7x - 7$	multiply by 7
$x(x - 1) = x^2 - x$	multiply by x, shift left
$x^2 + 6x - 7$	add the columns

Answer: $x^2 + 6x - 7$

3 Multiply bigger polynomials

1. Multiply the first term of the first bracket by every term of the second.
الحد الأول في كل حدود القوس الثاني.
2. Do the same for every other term (watch the signs).
كرر لباقي الحدود.
3. Combine like terms and write in standard form.
اجمع المتشابهات.

Find $(2x - 1)(x^2 + 4x - 3)$.

$2x(x^2 + 4x - 3) = 2x^3 + 8x^2 - 6x$	first term
$-1(x^2 + 4x - 3) = -x^2 - 4x + 3$	second term
$2x^3 + 7x^2 - 10x + 3$	combine

Answer: $2x^3 + 7x^2 - 10x + 3$

4 Area problems

1. Area of a rectangle = length × width.
المساحة = الطول × العرض.
2. If something extends **x** on every side, add **2x** to each dimension.
لو زاد x من كل ناحية زوّد 2x.
3. Multiply, then combine like terms.
اضرب ثم بسّط.

A rectangle has length $2x + 5$ and width $x + 3$. Find its area.

$A = (2x + 5)(x + 3)$	length × width
$= 2x^2 + 6x + 5x + 15$	FOIL
$= 2x^2 + 11x + 15$	combine

Answer: $2x^2 + 11x + 15$

DON'T FALL FOR IT

✗	✓	
$(x + 4)(x - 6) = x^2 - 24$	$x^2 - 2x - 24$	Do not forget the Outer and Inner products. متنساش الخارجي والداخلي.
$(3x - 4)(4x - 10) = 12x^2 - 46x - 40$	$+40$	Last: $(-4)(-10) = +40$. سالب × سالب = موجب.
FOIL for $(x + 3)(x^2 - x - 9)$	Distribute every term	FOIL only works for two binomials. Here there are 6 products. لقوسين فيهم حدين بس FOIL.

LAST LOOK · LESSON 11-3 · THE NIGHT BEFORE THE EXAM

Every Term Meets

Multiplying Polynomials

SCAN FOR PRACTICE

1 FOIL (2 binomials)

First, Outer, Inner, Last. Then combine the two middle terms.

$(x - 2)(x + 6) = x^2 + 4x - 12$

الأول، الخارجي، الداخلي، الأخير.

2 Vertical method

Write one binomial under the other and multiply like whole numbers.

x − 1 over x + 7

زي ضرب الأعداد تحت بعض.

Polynomial × Polynomial

3 Bigger polynomials

Multiply EVERY term of the first by EVERY term of the second, then combine.

2 terms × 3 terms = 6 products

كل حد في كل حد ثم اجمع.

4 Area

Area of a rectangle = length × width. Multiply the two expressions.

$(2x + 5)(x + 3)$

المساحة = الطول × العرض.

WORDS TO KNOW

binomial	ثنائية الحد
FOIL	طريقة FOIL
quadratic expression	مقدار تربيعي
Distributive Property	خاصية التوزيع

In the final exam: **Q2 · Q21** · scan the QR for timed practice

UNIT 1 · LESSON 11-4 · BOOK P.653

Shortcut Squares

Special Products

SCAN FOR PRACTICE

MUST KNOW

1 Square of a sum

1. Name **a** (first term) and **b** (second term).
حدد a و b.
2. Write $a^2 + 2ab + b^2$.
طبّق القانون.
3. Square everything inside **a** and **b**, including the numbers.
ربّع المعامل كمان.

Find $(3g + 10h)^2$.

$a = 3g, b = 10h$	name a and b
$(3g)^2 + 2(3g)(10h) + (10h)^2$	pattern
$9g^2 + 60gh + 100h^2$	simplify

Answer: $9g^2 + 60gh + 100h^2$

2 Square of a difference

1. Same as the sum, but the middle term is negative.
نفس القانون والحد الأوسط سالب.
2. The last term b^2 is always positive.
الحد الأخير موجب دايمًا.

Find $(7d - 2f)^2$.

$a = 7d, b = 2f$	name a and b
$(7d)^2 - 2(7d)(2f) + (2f)^2$	pattern
$49d^2 - 28df + 4f^2$	simplify

Answer: $49d^2 - 28df + 4f^2$

3 Sum × difference

1. Check: same two terms, one bracket with +, the other with −.
نفس الحدين، واحد + وواحد −.
2. Answer = $a^2 - b^2$ (no middle term).
الناتج $a^2 - b^2$ ومفيش حد أوسط.

Find $(3y^3 + 4)(3y^3 - 4)$.

$a = 3y^3, b = 4$	name a and b
$(3y^3)^2 - 4^2$	pattern
$9y^6 - 16$	power of a power: $(y^3)^2 = y^6$

Answer: $9y^6 - 16$

4 Real-life products

1. Find the formula: square $A = s^2$, cube surface area $A = 6s^2$.
حدد القانون.
2. Use the special product pattern, then multiply by any number in front.
طبّق القانون ثم اضرب في الرقم.

A cube has side $3n + 6$. Write its surface area $A = 6s^2$.

$(3n + 6)^2 = 9n^2 + 36n + 36$	square of a sum
$6(9n^2 + 36n + 36) = 54n^2 + 216n + 216$	times 6

Answer: $6(9n^2 + 36n + 36)$ or $54n^2 + 216n + 216$

DON'T FALL FOR IT

✗	✓	
$(x + 6)^2 = x^2 + 36$	$x^2 + 12x + 36$	Never forget the middle term **2ab**. متنساش الحد الأوسط.
$(2x + 9)^2 = 2x^2 + 36x + 81$	$4x^2 + 36x + 81$	Square the coefficient too: $(2x)^2 = 4x^2$. ربّع المعامل كمان.
$(4k - 1)^2 = 16k^2 - 8k - 1$	$+1$	b^2 is always positive. آخر حد موجب دايمًا.

LAST LOOK · LESSON 11-4 · THE NIGHT BEFORE THE EXAM

Shortcut Squares

Special Products

SCAN FOR PRACTICE

Special Products

1 Square of a sum

$(a + b)^2 = a^2 + 2ab + b^2$

$(x + 6)^2 = x^2 + 12x + 36$

مربع الأول + ٢×الأول×التاني + مربع التاني.

2 Square of a difference

$(a - b)^2 = a^2 - 2ab + b^2$. Only the middle sign is minus.

$(4k - 1)^2 = 16k^2 - 8k + 1$

نفس القانون والإشارة السالبة في النص بس.

3 Sum × difference

$(a + b)(a - b) = a^2 - b^2$. The middle terms cancel.

$(z + 5)(z - 5) = z^2 - 25$

مجموع × فرق = فرق مربعين.

4 Real-life

Area of a square $= s^2$, cube surface $= 6s^2$, Punnett square $(B + b)^2$.

$A = 6(3n + 6)^2$

مساحة المربع ومساحة سطح المكعب.

WORDS TO KNOW

square of a sum مربع مجموع	**square of a difference** مربع فرق	**product of a sum and a difference** حاصل ضرب مجموع في فرق	**perfect square trinomial** ثلاثية حدود مربع كامل
difference of squares فرق مربعين			

In the final exam: Q3 · scan the QR for timed practice

Pull It Out

Using the Distributive Property

SCAN FOR PRACTICE

MUST KNOW

1 Find the GCF

1. Find the greatest number that divides all the coefficients.
أكبر عدد بيقسم كل المعاملات.
2. For each variable in EVERY term, take its smallest exponent.
لكل متغير موجود في كل الحدود خد أصغر أس.
3. Multiply them: that is the GCF.
اضربهم.

Find the GCF of $20x^2y^2$, $-45x^2y$ and $-35x^2$.

Numbers: 20, 45, 35 → 5 — greatest common number

x: smallest power $x^2 \cdot y$ is not in $-35x^2$ — variables

GCF = $5x^2$ — multiply

Answer: $5x^2$

2 Factor out the GCF

1. Find the GCF.
أوجد العامل المشترك.
2. Divide each term by the GCF. Write the GCF outside the bracket.
اقسم كل حد عليه.
3. Multiply back to check.
اضرب للتأكد.

Factor $20x^2y^2 - 45x^2y - 35x^2$.

GCF = $5x^2$ — step 1

$5x^2(4y^2) + 5x^2(-9y) + 5x^2(-7)$ — divide each term

$5x^2(4y^2 - 9y - 7)$ — factored

Answer: $5x^2(4y^2 - 9y - 7)$

3 Real-life factoring

1. Write the expression from the question.
اكتب المقدار.
2. Factor out the GCF, often a number times **t**.
أخرج العامل المشترك.

A rock's height is $440t - 16t^2$. Factor the expression.

GCF of 440 and 16 is 8 · both terms have t — GCF = 8t

$8t(55) + 8t(-2t)$ — divide

$8t(55 - 2t)$ — factored

Answer: $8t(55 - 2t)$

DON'T FALL FOR IT

✗	✓	
$440t - 16t^2 = 4t(110 - 4t)$	$8t(55 - 2t)$	Factor COMPLETELY. 4 is not the greatest common factor. خد أكبر عامل مشترك.
$12a^2 + 16a = 4a(3a + 16)$	$4a(3a + 4)$	Divide EVERY term by the GCF. اقسم كل الحدود.
$6x^2 + 3x = 3x(2x)$	$3x(2x + 1)$	When a term equals the GCF, 1 is left in its place. لما الحد = العامل المشترك يتبقى 1.

Pull It Out

Using the Distributive Property

SCAN FOR PRACTICE

MUST KNOW

4 Group and factor

1. Group the first two terms and the last two terms.
جمّع أول حدين وآخر حدين.
2. Factor the GCF out of each group.
خذ العامل المشترك من كل مجموعة.
3. Factor out the common binomial.
طلّع القوس المشترك.

Factor $2uv + 6u + 5v + 15$.

$(2uv + 6u) + (5v + 15)$ — group

$2u(v + 3) + 5(v + 3)$ — GCF of each group

$(2u + 5)(v + 3)$ — common binomial

Answer: $(2u + 5)(v + 3)$

5 Additive inverses

1. Group and factor each pair.
جمّع وحلّل.
2. If you get $(7 - p)$ and $(p - 7)$, rewrite one: $7 - p = -1(p - 7)$.
اكتب واحد منهم بسالب.
3. Now factor out the common binomial.
طلّع القوس المشترك.

Factor $21m - 3mp + 4p - 28$.

$3m(7 - p) + 4(p - 7)$ — group and factor

$3m(-1)(p - 7) + 4(p - 7)$ — $7 - p = -(p - 7)$

$(-3m + 4)(p - 7)$ — common binomial

Answer: $(-3m + 4)(p - 7)$

DON'T FALL FOR IT

✗	✓	
$tw + 10t - 2w - 20 = t(w + 10) - 2($	$-2(w + 10)$	$-2 \cdot 10 = -20$. Check each group by multiplying back. راجع إشارة المجموعة التانية.
$(2u + 5)(v + 3)$ written as $(2u)(5)(v + 3)$	$(2u + 5)(v + 3)$	The leftover factors make ONE bracket: $2u + 5$. الباقي قوس واحد.
Grouping a trinomial $x^2 + 5x + 6$ directly	Needs 4 terms	Grouping needs four terms. A trinomial must be split first (next lesson). التجميع محتاج ٤ حدود.

LAST LOOK · LESSON 11-5 · THE NIGHT BEFORE THE EXAM

Pull It Out

Using the Distributive Property

SCAN FOR PRACTICE

Factor with the Distributive Property

1 Find the GCF

Biggest number that divides all coefficients × each shared variable with its smallest power.

$12a^2, 16a \rightarrow 4a$

أكبر عدد مشترك × المتغير بأصغر أس.

2 Factor it out

Divide every term by the GCF and write GCF(...).

$12a^2 + 16a = 4a(3a + 4)$

اقسم كل حد على العامل المشترك.

3 Factor completely

Check the bracket: no common factor may be left inside.

$8t(55 - 2t)$ ✓ $4t(110 - 4t)$ ×

اتأكد إن القوس مفيهوش عامل مشترك.

4 Check

Multiply back. You must get the original polynomial.

$4a(3a + 4) = 12a^2 + 16a$ ✓

اضرب تاني للتأكد.

5 When?

4 terms (or more) and no GCF for all of them.

$2uv + 6u + 5v + 15$

أربع حدود.

6 Group in pairs

Bracket the first two and the last two terms. Factor the GCF of each pair.

$2u(v + 3) + 5(v + 3)$

جمّع كل حدين وخد العامل المشترك.

7 Common binomial

The same bracket appears twice. Factor it out.

$(2u + 5)(v + 3)$

القوس المتكرر يطلع برا.

8 Additive inverse

If the brackets are opposites, take out a negative: $7 - p = -(p - 7)$.

$3m(7 - p) + 4(p - 7)$

لو القوسين عكس بعض طلّع سالب.

WORDS TO KNOW

factoring التحليل إلى عوامل	**greatest common factor (GCF)** العامل المشترك الأكبر	**factored completely** محلل تحليلًا كاملًا	**prime factorization** التحليل لعوامل أولية
factoring by grouping التحليل بالتجميع	**common binomial** ذات حدين مشترك	**additive inverses** معكوسات جمعية	

In the final exam: **Q4** · scan the QR for timed practice

UNIT 1 · LESSON 11-6 · BOOK P.667

Find the Pair

Factoring Quadratic Trinomials

SCAN FOR PRACTICE

MUST KNOW

1 c is positive

1. List factor pairs of **c** that have the same sign as **b**.
اكتب أزواج عوامل c بنفس إشارة b.
2. Pick the pair whose sum is **b**.
اختار الزوج اللي مجموعه b.
3. Write **(x + m)(x + p)** and check with FOIL.
اكتب الناتج وراجع.

Factor $x^2 - 9x + 18$.

m = −3, p = −6

product 18, sum −9

(x − 3)(x − 6)

write the factors

Answer: (x − 3)(x − 6)

Factors of 18	Sum
−1, −18	−19
−2, −9	−11
−3, −6	−9 ✓

2 c is negative

1. The two numbers have opposite signs.
الرقمين إشارتهم مختلفة.
2. The number with the bigger size takes the sign of **b**.
الأكبر قيمة ياخد إشارة b.

Factor $x^2 + 5x - 14$.

Pairs of −14: (−1, 14) sum 13 · (−2, 7) sum 5 ✓

product −14, sum 5

(x − 2)(x + 7)

write the factors

Answer: (x − 2)(x + 7)

3 Leading coefficient ≠ 1

1. Take out a GCF if there is one.
خد العامل المشترك لو موجود.
2. Find **m** and **p** with **m + p = b** and **mp = ac**.
رقمين مجموعهم b وضربهم ac.
3. Split **bx** into **mx + px**, then factor by grouping.
قسّم الحد الأوسط وحلّل بالتجميع.

Factor $2x^2 - 17x + 21$.

ac = 2 · 21 = 42. Need product 42, sum −17 → −3 and −14

find m and p

$2x^2 - 3x - 14x + 21$

split the middle term

x(2x − 3) − 7(2x − 3) = (2x − 3)(x − 7)

group

Answer: (2x − 3)(x − 7)

4 Prime trinomials and problems

1. If no pair of integers works, write **prime**.
لو مفيش زوج ينفع اكتب أولية.
2. In word problems: factor the area to find the sides, then change them as the question says.
في المسائل حلّل المساحة لتعرف الأبعاد.

A square flag has area $x^2 - 6x + 9$. The length is increased by 4 ft. Find the new area.

$x^2 - 6x + 9 = (x - 3)(x - 3)$

side = x − 3

New: (x − 3 + 4)(x − 3) = (x + 1)(x − 3)

add 4 to the length

$= x^2 - 2x - 3$

FOIL

Answer: $x^2 - 2x - 3$ ft²

DON'T FALL FOR IT

✗ $x^2 + 5x - 14 = (x + 2)(x - 7)$	✓ $(x - 2)(x + 7)$	Check the sum: **2 − 7 = −5**, not 5. The bigger number takes the sign of **b**. الأكبر ياخد إشارة b.
✗ $4x^2 + 18x - 10 = (2x - 1)(2x + 10)$	✓ $2(2x - 1)(x + 5)$	Take out the GCF first and factor completely. العامل المشترك الأول.
✗ $2x^2 - 17x + 21$: numbers with product 21	✓ product **ac = 42**	When **a ≠ 1**, use **ac**, not **c**. لما a ≠ 1 استخدم ac.

LAST LOOK · LESSON 11-6 · THE NIGHT BEFORE THE EXAM

Find the Pair

Factoring Quadratic Trinomials

SCAN FOR PRACTICE

Factor

1 c positive

Both numbers have the SAME sign as **b**.

$x^2 - 9x + 18 = (x - 3)(x - 6)$

c موجب: الرقمين نفس إشارة b.

2 c negative

The numbers have OPPOSITE signs. The bigger one takes the sign of **b**.

$x^2 + 5x - 14 = (x - 2)(x + 7)$

c سالب: إشارتين مختلفتين، والأكبر بإشارة b.

3 a ≠ 1

Find two numbers with product **ac** and sum **b**. Split the middle term, then group.

$2x^2 - 17x + 21 = (2x - 3)(x - 7)$

حاصل الضرب ac والمجموع b، ثم التجميع.

4 GCF first · prime

Always take out the GCF first. No pair works → the trinomial is prime.

$x^2 - 4x + 8$ **is prime**

العامل المشترك أولًا، ولو مفيش رقمين تبقى أولية.

WORDS TO KNOW

quadratic trinomial	ثلاثية حدود تربيعية
leading coefficient	المعامل الرئيسي
prime polynomial	كثيرة حدود أولية
factor pair	زوج عوامل

In the final exam: **Q5 · Q22** · scan the QR for timed practice

UNIT 1 · LESSON 11-7 · BOOK P.675

Spot the Square

Factoring Special Products

SCAN FOR PRACTICE

MUST KNOW

1 Difference of squares

1. Check: two terms, both perfect squares, minus between them.
   حدين، مربعات كاملة، وسالب بينهم.
2. Write each as a square: a^2 and b^2.
   اكتب كل حد كمربع.
3. Answer: $(a + b)(a - b)$.
   الناتج.

Factor $81v^2 - 64w^2$.

$(9v)^2 - (8w)^2$ — write as squares

$(9v + 8w)(9v - 8w)$ — pattern

Answer: $(9v + 8w)(9v - 8w)$

2 Factor more than once

1. Take out any GCF first.
   العامل المشترك أولًا.
2. Factor the difference of squares.
   حلّل فرق المربعين.
3. Check each bracket: a difference of squares again? Factor again. A sum of squares stays.
   لو قوس لسه فرق مربعين حلّله تاني.

Factor $x^4 - 256$.

$(x^2)^2 - 16^2 = (x^2 + 16)(x^2 - 16)$ — first time

$x^2 - 16 = (x + 4)(x - 4)$ — again

$(x^2 + 16)(x + 4)(x - 4)$ — $x^2 + 16$ is prime

Answer: $(x^2 + 16)(x + 4)(x - 4)$

3 Perfect square trinomials

1. Is the first term a perfect square? Is the last term a perfect square (and positive)?
   الأول والأخير مربعات؟
2. Is the middle term $2 \cdot a \cdot b$?
   الأوسط = 2ab؟
3. If yes: $(a + b)^2$ or $(a - b)^2$ (sign of the middle term).
   لو آه: مربع مجموع أو فرق.

Is $36h^2 - 12h + 1$ a perfect square trinomial? If so, factor it.

$36h^2 = (6h)^2$ ✓ · $1 = 1^2$ ✓ — first and last

$2(6h)(1) = 12h$ ✓ — middle

$(6h - 1)^2$ — factor

Answer: yes · $(6h - 1)^2$

4 Area problems

1. Area left over = big area − small area.
   المساحة الباقية = الكبيرة − الصغيرة.
2. This is often a difference of squares: factor it.
   غالبًا فرق مربعين.

A 6 m by 6 m room has an x m by x m tiled square. Write the carpeted area in factored form.

$36 - x^2$ — big square − small square

$(6 + x)(6 - x)$ — difference of squares

Answer: $(6 + x)(6 - x)$ m²

DON'T FALL FOR IT

✗ $x^2 + 16 = (x + 4)(x - 4)$	✓ prime	$(x + 4)(x - 4) = x^2 - 16$. A SUM of squares cannot be factored. مجموع مربعين أولي.
✗ $x^4 - 256 = (x^2 + 16)(x^2 - 16)$ (stop)	✓ $(x^2 + 16)(x + 4)(x - 4)$	$x^2 - 16$ is still a difference of squares. Factor completely. كمّل التحليل.
✗ $36x^2 - 36x + 9 = (6x - 3)^2$	✓ $9(2x - 1)^2$	Take out the GCF first: 9. خد العامل المشترك الأول.

LAST LOOK · LESSON 11-7 · THE NIGHT BEFORE THE EXAM

Spot the Square

Factoring Special Products

SCAN FOR PRACTICE

Factor special products

1 Difference of squares

Two perfect squares with a minus between them.

$x^2 - 9 = (x + 3)(x - 3)$

مربعين بينهم سالب.

2 Factor more than once

Take out the GCF first. If a bracket is still a difference of squares, factor again.

$x^4 - 256 = (x^2 + 16)(x + 4)(x - 4)$

العامل المشترك أولًا، وحلّل تاني لو ينفع.

3 Perfect square trinomial

First and last are squares, middle = **2ab**.

$x^2 + 10x + 25 = (x + 5)^2$

الأول والأخير مربعات والأوسط 2ab.

4 Sum of squares is prime

$a^2 + b^2$ cannot be factored.

$x^2 + 16$ is prime

مجموع مربعين لا يتحلل.

WORDS TO KNOW

difference of two squares فرق بين مربعين	**perfect square** مربع كامل	**perfect square trinomial** ثلاثية حدود مربع كامل	**factor completely** التحليل الكامل

In the final exam: **Q6** · scan the QR for timed practice

UNIT 2 · LESSON 12-1 · PART A · BOOK P.689

The Turning Point

Graphing Quadratic Functions

SCAN FOR PRACTICE

MUST KNOW

1 Read the graph

1. Vertex = the turning point. Axis = the vertical line through it: **x = h**.
الرأس ونقطة الدوران.
2. **y**-intercept = where it crosses the **y**-axis.
نقطة التقاطع مع محور y.
3. End behavior: opens down → as **x** increases or decreases, **y** decreases.
السلوك الطرفي.

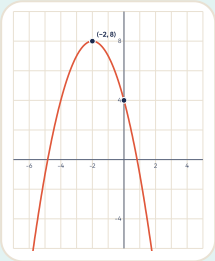
Identify the axis of symmetry, vertex, y-intercept and end behavior.

Axis x = −2 · vertex (−2, 8) (maximum) — turning point

y-intercept 4 — crosses the y-axis at (0, 4)

Opens down: as x increases or decreases, y decreases — end behavior

Answer: x = −2, (−2, 8), 4, y decreases both ways



2 Axis and vertex from the equation

1. Read **a**, **b**, **c**.
حدد a و b و c.
2. Axis: $x = -\frac{b}{2a}$.
محور التماثل.
3. Substitute into **f(x)** for the **y** of the vertex.
عوّض لإيجاد y.

Find the axis of symmetry and vertex of $f(x) = x^2 - 4x + 1$.

$x = -\frac{-4}{2(1)} = 2$ — axis

$f(2) = 4 - 8 + 1 = -3$ — substitute

vertex (2, −3) — minimum since a > 0

Answer: x = 2, vertex (2, −3)

3 Maximum or minimum value

1. Look at **a**: positive → minimum, negative → maximum.
إشارة a.
2. Find the vertex.
أوجد الرأس.
3. The max/min VALUE is the **y**-coordinate of the vertex. Range: **y ≥** min or **y ≤** max.
القيمة = y للرأس.

Does $f(x) = -x^2 - 2x + 2$ have a maximum or minimum? Find it, the domain and the range.

a = −1 < 0 → maximum — opens down

$x = -\frac{-2}{2(-1)} = -1$, $f(-1) = -1 + 2 + 2 = 3$ — vertex

Domain: all real numbers · Range: y ≤ 3 — domain, range

Answer: maximum value 3

4 Graph with a table

1. Find the axis and vertex.
أوجد المحور والرأس.
2. Make a table with **x** values on both sides of the vertex.
جدول قيم حول الرأس.
3. Plot the points and draw a smooth U-shape.
ارسم المنحنى.

Graph $y = 2x^2 - 8x + 2$.

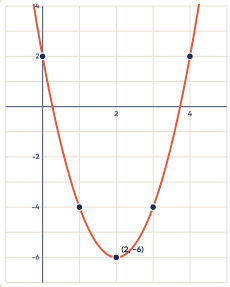
$x = -\frac{-8}{2(2)} = 2$ — axis

x	0	1	2	3	4
y	2	−4	−6	−4	2

table

Vertex (2, −6) minimum, y-intercept 2 — key points

Graph:



DON'T FALL FOR IT

✗	✓	
Axis $x = \frac{b}{2a}$	$x = -\frac{b}{2a}$	Do not forget the minus sign. متنساش السالب.
a < 0 → minimum	maximum	Opens down (sad face) → highest point → maximum. a سالب يبقى عظمى.
Maximum value of $-(x-3)^2 + 5$ is 3	5	The max/min VALUE is the **y** of the vertex, not the **x**. القيمة هي y مش x.

UNIT 2 · LESSON 12-1 · PART B · BOOK P.693

The Turning Point

Graphing Quadratic Functions

SCAN FOR PRACTICE

MUST KNOW

5 Maximum height

1. $t = -\frac{b}{2a}$ gives the time of the maximum.

زمن أقصى ارتفاع.

2. Substitute that **t** to get the maximum height.

عوّض.

3. Write the answer with units.

اكتب الوحدة.

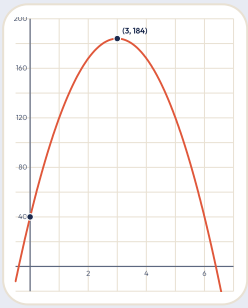
A catapult's boulder has height $h(t) = -16t^2 + 96t + 40$ (feet). Find the maximum height and when it happens.

$t = -\frac{96}{2(-16)} = 3$ time of max

$h(3) = -16(9) + 96(3) + 40 = 184$ substitute

Starts at $h(0) = 40$ ft y-intercept meaning

Answer: 184 ft after 3 seconds



6 Greatest area · greatest income

1. Write the quantity as a quadratic:
$A = x(\text{half perimeter} - x)$, R = price × number sold.

اكتب الدالة.

2. **$a < 0$**, so the vertex gives the greatest value.

الرأس يعطي أكبر قيمة.

40 m of fence makes a rectangle. Find the greatest area.

Width x, length 20 − x half the perimeter = 20

$A = x(20 - x) = -x^2 + 20x$ area function

$x = -\frac{20}{-2} = 10$, $A = 10 \cdot 10 = 100$ vertex

Answer: 100 m² (a 10 m × 10 m square)

7 Interpret key features

1. **y**-intercept → start value. Vertex → max/min. **x**-intercepts → when it is 0.

معنى كل نقطة.

2. Increasing before the vertex, decreasing after it (for a maximum).

متزايدة قبل الرأس ومتناقصة بعده.

3. Domain: only times that make sense (from 0 until it lands).

المجال المنطقي.

For $h(t) = -16t^2 + 96t + 40$, the ball lands at about $t = 6.4$. Describe the key features.

Increasing for $t < 3$, decreasing for $t > 3$ before/after the vertex

Domain $0 \le t \le 6.4$ · range $0 \le h \le 184$ sensible values

Answer: rises for 3 s to 184 ft, lands at ≈ 6.4 s

DON'T FALL FOR IT

× Max height of $-16t^2 + 96t + 40$ is 3	✓ 184 ft	3 is the TIME. Substitute it to get the height. ٣ هو الزمن مش الارتفاع.
× Domain: all real numbers	✓ $0 \le t \le$ landing time	Time cannot be negative, and the ball stops at the ground. الزمن لا يكون سالب.
× $t = -\frac{96}{2(16)}$	✓ $-\frac{96}{2(-16)} = 3$	**a** is −16. Keep its sign. a بإشارتها.

LAST LOOK · LESSON 12-1 · THE NIGHT BEFORE THE EXAM

The Turning Point

Graphing Quadratic Functions

SCAN FOR PRACTICE

1 Axis of symmetry

$x = -\frac{b}{2a}$. The vertical line through the vertex.

$y = x^2 + 2x - 6 \rightarrow x = -1$

محور التماثل.

2 Vertex

Put the axis value into **f(x)** to get **y**.

$f(-1) = -7 \rightarrow (-1, -7)$

الرأس: عوّض في الدالة.

3 Max or min

a > 0 opens up → minimum. **a < 0** opens down → maximum. The value is the **y** of the vertex.

$a = 1 > 0 \rightarrow$ min -7

a موجب صغرى، سالب عظمى.

4 Graph it

Axis, vertex, **y**-intercept **(0, c)**, a table of points on both sides, connect smoothly.

y-intercept = c

ارسم بجدول قيم حول الرأس.

Parabola

5 Height of an object

$h(t) = -16t^2 + vt + h_0$ (feet) or $-5t^2$ (metres). **a < 0** → maximum height.

$h(t) = -16t^2 + 96t + 40$

الارتفاع: دالة عظمى.

6 When is the max?

Time $t = -\frac{b}{2a}$ (the axis of symmetry).

$t = 3$ s

الزمن = محور التماثل.

7 What is the max?

Substitute that time: **h(t)** is the maximum height.

$h(3) = 184$ ft

عوّض لإيجاد أقصى ارتفاع.

8 Read the meaning

y-intercept = starting height. **x**-intercept = hits the ground. Domain = sensible times only.

$h(0) = 40$ ft

معنى كل نقطة في السؤال.

WORDS TO KNOW

quadratic function دالة تربيعية	**parabola** قطع مكافئ	**axis of symmetry** محور التماثل	**vertex** الرأس
maximum · minimum قيمة عظمى · صغرى	**end behavior** السلوك الطرفي	**domain · range** المجال · المدى	**maximum height** أقصى ارتفاع
initial height الارتفاع الابتدائي	**zeros** الأصفار	**reasonable domain** مجال منطقي	

In the final exam: **Q7 · Q23** · scan the QR for timed practice

UNIT 2 · LESSON 12-2 · BOOK P.699

Move the Parabola

Transformations of Quadratic Functions

SCAN FOR PRACTICE

MUST KNOW

1 Translations

1. Find **k** (outside): up if positive, down if negative.
k: فوق أو تحت.
2. Find **h** (inside): **(x − h)**. **(x − 3)** → right 3; **(x + 3)** → left 3.
h: عكس الإشارة.
3. Vertex of **g** is **(h, k)**.
الرأس (h, k).

Describe $g(x) = (x - 3)^2 + 1$.

h = 3 → right 3 — inside

k = 1 → up 1 — outside

vertex (3, 1)

Answer: translated 3 units right and 1 unit up

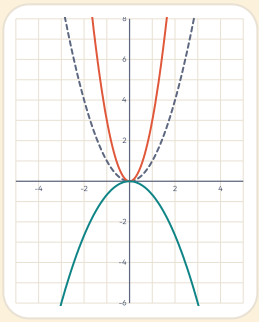
2 Dilations

1. Look at **|a|**.
انظر لـ |a|.
2. **|a| > 1**: vertical stretch (narrower).
أكبر من 1: تمدد رأسي (أضيق).
3. **0 < |a| < 1**: vertical compression (wider).
بين 0 و1: انكماش (أوسع).

Describe $g(x) = 3x^2$ and $h(x) = -\frac{2}{3}x^2$.

g: a = 3 > 1 → stretched vertically — narrower

h: $|a| = \frac{2}{3} < 1$ → compressed, and a < 0 → reflected — wider, opens down



Answer: g stretched · h compressed and reflected across the x-axis

3 Reflections and combined

1. Negative **a** → reflected across the **x**-axis.
a سالب = انعكاس.
2. Then describe **|a|**, **h** and **k** in turn.
ثم باقي التحويلات.

Describe $g(x) = -2(x - 3)^2 + 8$.

a = −2: reflected, stretched — shape

h = 3 right · k = 8 up — position

vertex (3, 8), maximum — opens down

Answer: reflected across the x-axis, stretched, 3 right, 8 up

4 Equation from a graph

1. Read the vertex **(h, k)**.
اقرأ الرأس.
2. Opens up → **a > 0**; down → **a < 0**. Same width as x^2 → **a = ± 1**.
اتجاه الفتحة.
3. Write $y = a(x - h)^2 + k$.
اكتب المعادلة.

A parabola opens up, has the same shape as $y = x^2$ and its vertex is (3, 1). Write its equation.

a = 1, h = 3, k = 1 — read

$y = (x - 3)^2 + 1$ — vertex form

Answer: $y = (x - 3)^2 + 1$

DON'T FALL FOR IT

× $(x + 2)^2$ → right 2	✓ left 2	Inside the bracket the direction is opposite to the sign. جوه القوس عكس الإشارة.
× $\frac{1}{2}x^2$ is stretched	✓ compressed	**0 < |a| < 1** makes it wider: a vertical compression. أقل من 1 = انكماش.
× Vertex of $(x - 4)^2 + 2$ is **(−4, 2)**	✓ **(4, 2)**	In $a(x - h)^2 + k$, **h = 4**. h = 4.

LAST LOOK · LESSON 12-2 · THE NIGHT BEFORE THE EXAM

Move the Parabola

Transformations of Quadratic Functions

SCAN FOR PRACTICE

1 k: up or down

k > 0 up, **k < 0** down.

$x^2 - 4$ → down 4

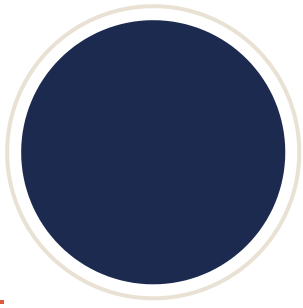
k برا القوس: فوق أو تحت.

2 h: right or left

Inside the bracket the sign is OPPOSITE:
(x − 3) → right 3, **(x + 2)** → left 2.

$(x + 2)^2$ → left 2

h جوه القوس بعكس الإشارة.



3 a: stretch or compress

|a| > 1 stretched (narrower). **0 < |a| < 1** compressed (wider).

$3x^2$ stretched · $\frac{1}{2}x^2$ compressed

a أكبر من 1 تضييق، أقل من 1 توسيع.

4 a < 0: reflect

A negative **a** reflects the graph across the **x**-axis (opens down).

$-x^2$ opens down

a سالب = انعكاس حول محور x.

WORDS TO KNOW

parent function الدالة الأم	**translation** انسحاب	**dilation** تمدد / انكماش	**reflection** انعكاس
vertex form الصورة الرأسية			

In the final exam: **Q8** · scan the QR for timed practice

UNIT 2 · LESSON 12-3 · PART A · BOOK P.711

Where It Lands

Solving Quadratic Equations by Graphing

SCAN FOR PRACTICE

MUST KNOW

1 Two roots

1. Write the equation as $ax^2 + bx + c = 0$.
الصيغة القياسية.
2. Graph the related function (axis, vertex, table).
ارسم الدالة.
3. Read the **x**-intercepts. Check by substituting.
اقرأ نقاط التقاطع وتحقق.

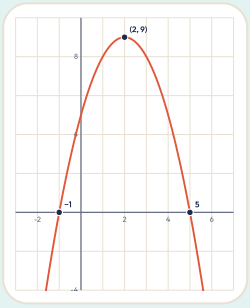
Solve $-x^2 + 4x + 5 = 0$ by graphing.

Vertex: x = 2, f(2) = 9
axis and vertex

x-intercepts at −1 and 5 read

Check: $-(5)^2 + 4(5) + 5 = 0$ ✓
substitute

Answer: x = −1 and x = 5



2 One double root

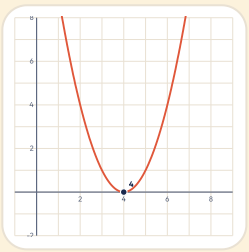
1. Rewrite with 0 on one side.
خلّي الطرف التاني صفر.
2. If the vertex lies ON the **x**-axis, there is only one solution.
لو الرأس على المحور يبقى حل واحد.

Solve $x^2 - 8x = -16$ by graphing.

$x^2 - 8x + 16 = 0$ standard form

Vertex (4, 0) is on the x-axis
one intercept

Answer: x = 4 (double root)



3 No real roots

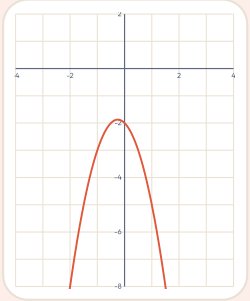
1. Graph the related function.
ارسم الدالة.
2. Opens down with vertex below the axis, or opens up with vertex above it → no intercepts.
الرأس والفتحة في عكس اتجاه المحور.
3. Write: no real solutions.
اكتب: لا يوجد حلول حقيقية.

Solve $-2x^2 - x - 2 = 0$ by graphing.

Opens down, vertex below the x-axis
f(−0.25) ≈ −1.9

No x-intercepts

Answer: no real solutions (ø)



DON'T FALL FOR IT

✗ Solve $x^2 - 8x = -16$ by reading where $y = x^2 - 8x$ crosses the axis	✓ graph $x^2 - 8x + 16$	Get 0 on one side first. الطرف التاني لازم يبقى صفر.
✗ The solutions are the **y**-intercept	✓ the **x**-intercepts	Solutions are where **f(x) = 0**, on the **x**-axis. الحلول على محور x.
✗ No **x**-intercepts → **x = 0**	✓ no real solutions	0 is a number. Here there is no real number that works. لا يوجد حل ، مش الحل صفر.

Where It Lands

Solving Quadratic Equations by Graphing

SCAN FOR PRACTICE

MUST KNOW

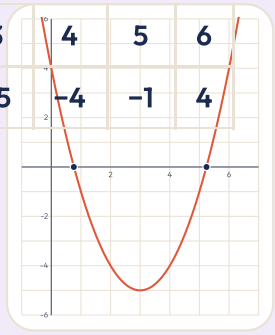
4 Between which integers?

1. Make a table of integer values of **x**.
جدول بأعداد صحيحة.
2. Look for a sign change in **f(x)**.
دوّر على تغيّر الإشارة.
3. The root lies between those two integers.
الجذر بينهم.

Between which integers are the roots of $x^2 - 6x + 4 = 0$?

x	0	1	2	3	4	5	6
f(x)	4	−1	−4	−5	−4	−1	4

table



Sign changes: 0 → 1 and 5 → 6

Answer: between 0 and 1, and between 5 and 6

5 Nearest tenth

1. Find the two integers around the root.
حدد العددين.
2. Table with steps of 0.1 between them.
جدول بخطوة 0.1.
3. Choose the **x** where **f(x)** is closest to 0.
الأقرب للصفر.

Estimate the greater root of $x^2 - 6x + 4 = 0$ to the nearest tenth.

f(5.2) = −0.16 · f(5.3) = 0.29 table

−0.16 is closer to 0 closest

Answer: x ≈ 5.2

6 Real-life estimates

1. Set the height equal to 0.
خلّي الارتفاع صفر.
2. Estimate the positive root (time cannot be negative).
الجذر الموجب.

A baseball's height is $h = -16t^2 + 105t + 1.5$. About how long is it in the air?

h(6.5) = 8 · h(6.6) = −2.46 sign change

−2.46 is closest to 0

Answer: about 6.6 seconds

DON'T FALL FOR IT

✗ **f(5) = −1, f(6) = 4** → root = **5**	✓ between 5 and 6	A root is where **f = 0**; the sign change only tells you the interval. تغيّر الإشارة يحدد الفترة بس.
✗ Baseball time ≈ **−0.01** s	✓ ≈ 6.6 s	Time cannot be negative. Use the sensible root. الزمن موجب.

LAST LOOK · LESSON 12-3 · THE NIGHT BEFORE THE EXAM

Where It Lands

Solving Quadratic Equations by Graphing

SCAN FOR PRACTICE

1 Standard form first

Move everything to one side:
$ax^2 + bx + c = 0$. Graph $f(x) = ax^2 + bx + c$.

$x^2 - 8x = -16 \rightarrow x^2 - 8x + 16 = 0$

خلّي الطرف التاني صفر.

2 Two roots

The graph crosses the **x**-axis twice.

$-x^2 + 4x + 5 = 0 \rightarrow -1, 5$

يقطع محور x مرتين.

3 One double root

The vertex sits on the **x**-axis.

$x^2 - 8x + 16 = 0 \rightarrow 4$

الرأس على محور x.

4 No real roots

The graph never touches the **x**-axis.

$-2x^2 - x - 2 = 0 \rightarrow$ **no real solutions**

لا يقطع محور x.

Roots = x-intercepts

5 Between integers

Find two consecutive integers where **f(x)** changes sign. A root is between them.

f(5) = −1, f(6) = 4 → between 5 and 6

الإشارة بتتغير يبقى فيه جذر.

6 To the nearest tenth

Make a table between those integers in steps of 0.1. Pick the value of **f** closest to 0.

≈ 5.2

جدول بخطوة 0.1.

7 Real-life

"When does it hit the ground?" means **h(t) = 0**. Keep only the sensible (positive) root.

t ≈ 6.6 s

الوصول للأرض: h = 0 والزمن موجب.

8 Calculator

Graph it and use the zero feature. Round as the question asks.

zero ≈ 0.76

الآلة الحاسبة.

WORDS TO KNOW

quadratic equation معادلة تربيعية	**root · solution** جذر · حل	**zero of a function** صفر الدالة	**double root** جذر مكرر
estimate تقدير	**sign change** تغيّر الإشارة	**nearest tenth** لأقرب جزء من عشرة	

In the final exam: **Q9** · scan the QR for timed practice

UNIT 2 · LESSON 12-4 · PART A · BOOK P.715, P.716

Zero Wins

Solving Quadratic Equations by Factoring

SCAN FOR PRACTICE

MUST KNOW

1 Square Root Property

1. Get the square alone: **(\ldots)² = n**.
خلّي المربع لوحده.
2. Take the square root of both sides: **±√n**.
جذر الطرفين بـ ±.
3. Solve both equations.
حل الاتنين.

Solve $(y + 5)^2 = 21$.

$y + 5 = \pm\sqrt{21}$ — square root

$y = -5 \pm \sqrt{21}$ — subtract 5

$y \approx -0.42$ **or** $y \approx -9.58$ — calculator

Answer: $y = -5 \pm \sqrt{21}$

2 Solve a factored equation

1. The equation must be (factor)(factor) = 0.
لازم الطرف التاني صفر.
2. Set each factor equal to 0.
كل قوس = صفر.
3. Solve each small equation.
حل كل واحدة.

Solve $(7x + 3)(2x - 6) = 0$.

$7x + 3 = 0 \rightarrow x = -\frac{3}{7}$ — first factor

$2x - 6 = 0 \rightarrow x = 3$ — second factor

Answer: $x = -\frac{3}{7}$ **or** $x = 3$

3 GCF then zero product

1. Move everything to one side (= 0).
خلّي الطرف التاني صفر.
2. Factor out the GCF.
أخرج العامل المشترك.
3. Set each factor to 0. Never divide by the variable.
كل عامل = صفر.

Solve $4x^2 - 12x = 0$.

$4x(x - 3) = 0$ — factor the GCF

$4x = 0 \rightarrow x = 0 \cdot x - 3 = 0 \rightarrow x = 3$ — zero product

Check: $4(0)^2 - 12(0) = 0$ ✓, $4(9) - 36 = 0$ ✓

Answer: $x = 0$ **or** $x = 3$

4 Write a function from its zeros

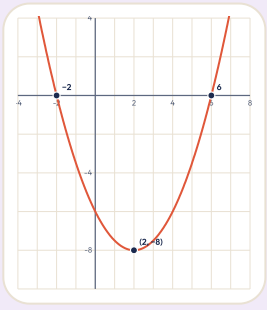
1. Zeros **p** and **q** → factors **(x − p)** and **(x − q)**.
الأصفار → العوامل.
2. Write **f(x) = a(x − p)(x − q)**.
اكتب الدالة.
3. Substitute another point to find **a**.
عوّض بنقطة لإيجاد a.

Write a quadratic function for the graph.

Zeros −2 and 6 →
$f(x) = a(x + 2)(x - 6)$ — factors

Point (2, −8):
$-8 = a(4)(-4) = -16a$ — substitute

$a = \frac{1}{2} \rightarrow f(x) = \frac{1}{2}x^2 - 2x - 6$



Answer:
$f(x) = \frac{1}{2}(x + 2)(x - 6)$

DON'T FALL FOR IT

× $(x - 2)^2 = 16 \rightarrow x = 6$	✓ $x = 6$ or $x = -2$	The square root gives ±. متنساش ±.
× $4x^2 = 12x$ → divide by x → $x = 3$	✓ $x = 0$ or $x = 3$	Dividing by **x** loses the root 0. Factor instead. القسمة على x بتضيّع الحل صفر.
× $(x + 2)(x - 4) = 0 \rightarrow x = 2, -4$	✓ $x = -2, 4$	Solve $x + 2 = 0$: $x = -2$. الإشارة بتتعكس.

UNIT 2 · LESSON 12-4 · PART B · BOOK P.717

Zero Wins

Solving Quadratic Equations by Factoring

SCAN FOR PRACTICE

MUST KNOW

5 Trinomials

1. Write in standard form (= 0).
الصيغة القياسية.
2. Factor the trinomial.
حلّل.
3. Set each factor to 0 and check.
كل عامل = صفر.

Solve $x^2 - 6x = 27$.

$x^2 - 6x - 27 = 0$ — standard form

$(x - 9)(x + 3) = 0$ — factor: $-9 \times 3 = -27$, $-9 + 3 = -6$

$x = 9$ or $x = -3$ — zero product

Answer: $x = 9$ or $x = -3$

6 Difference of squares

1. Write it as $a^2 - b^2 = 0$.
اكتبها فرق مربعين.
2. $(a + b)(a - b) = 0$.
حلّل.
3. Two roots: ±.
حلين موجب وسالب.

Solve $49 - x^2 = 0$.

$7^2 - x^2 = (7 + x)(7 - x) = 0$ — difference of squares

$x = -7$ or $x = 7$ — zero product

Answer: $x = \pm 7$

7 Perfect square trinomials

1. Get 0 on one side.
صفر في طرف.
2. Factor as $(a \pm b)^2$.
حلّل كمربع.
3. One (double) root.
جذر واحد مكرر.

Solve $25y^2 - 60y + 81 = 45$.

$25y^2 - 60y + 36 = 0$ — subtract 45

$(5y - 6)^2 = 0$ — perfect square

$5y - 6 = 0 \rightarrow y = \frac{6}{5}$

Answer: $y = \frac{6}{5}$

8 Word problems

1. Name the unknown and write the equation.
سمّ المجهول واكتب المعادلة.
2. Solve by factoring.
حل بالتحليل.
3. Reject answers that make no sense (negative lengths, negative time).
ارفض السالب.

An isosceles triangle has area 108 ft², legs 15 ft, base $4x + 2$ and height $3x$. Find its perimeter.

$108 = \frac{1}{2}(4x + 2)(3x) = 6x^2 + 3x$ — area

$6x^2 + 3x - 108 = 0 \rightarrow 3(2x + 9)(x - 4) = 0$ — factor

$x = 4$ (not $-\frac{9}{2}$) → base 18 — reject negative

Perimeter = 15 + 15 + 18 = 48

Answer: 48 ft

DON'T FALL FOR IT

✗ $x^2 - 6x = 27 \rightarrow x(x - 6) = 27 \rightarrow x = 27$...	✓ $x^2 - 6x - 27 = 0$ first	The product must equal 0 before you use the Zero Product Property. لازم = صفر.
✗ $49 - x^2 = 0 \rightarrow x = 7$	✓ $x = \pm 7$	A difference of squares gives two roots. حلين.
✗ Triangle: $x = -\frac{9}{2}$	✓ reject it	A length cannot be negative. الطول لا يكون سالب.

LAST LOOK · LESSON 12-4 · THE NIGHT BEFORE THE EXAM

Zero Wins

Solving Quadratic Equations by Factoring

SCAN FOR PRACTICE

Solve by factoring

1 Square Root Property

If $x^2 = n$ then $x = \pm\sqrt{n}$. Always two signs.

$(y + 5)^2 = 21 \rightarrow y = -5 \pm \sqrt{21}$

الجذر بـ ±.

2 Zero Product Property

If a product is 0, at least one factor is 0. Set EACH factor equal to 0.

$(x + 2)(x - 4) = 0 \rightarrow -2, 4$

كل عامل = صفر.

3 Factor out the GCF

$4x^2 - 12x = 0 \rightarrow 4x(x - 3) = 0$. Do not divide by x.

$x = 0$ or $x = 3$

العامل المشترك ومتقسمش على x.

4 Check

Substitute each root into the original equation.

$4(3)^2 - 12(3) = 0$ ✓

عوّض للتحقق.

5 Zeros → function

Zeros p and q give $f(x) = a(x - p)(x - q)$. Use another point to find a.

zeros $-2, 6 \rightarrow a(x + 2)(x - 6)$

من الأصفار للدالة.

6 Standard form

Move every term to one side: $ax^2 + bx + c = 0$.

$x^2 - 6x = 27 \rightarrow x^2 - 6x - 27 = 0$

خلّي الطرف التاني صفر.

7 Factor

Use the right method: GCF, trinomial, difference of squares, perfect square.

$(x - 9)(x + 3) = 0$

اختار طريقة التحليل.

8 Word problems

Write the equation, solve, then keep only answers that make sense.

consecutive integers $n, n + 1$

ارفض الإجابات غير المنطقية.

WORDS TO KNOW

Square Root Property خاصية الجذر التربيعي	**Zero Product Property** خاصية الضرب الصفري	**roots** جذور	**zeros** أصفار الدالة
factored form الصورة التحليلية	**standard form of an equation** الصيغة القياسية للمعادلة	**consecutive integers** أعداد صحيحة متتالية	**double root** جذر مكرر

In the final exam: **Q10** · scan the QR for timed practice

Make It Square

Solving Quadratic Equations by Completing the Square

SCAN FOR PRACTICE

MUST KNOW

1 Find c for a perfect square

1. Take half of **b**.

نص b.

2. Square it. That is **c**.

ربّعه.

Find c so that $x^2 + 8x + c$ is a perfect square trinomial.

$\frac{8}{2} = 4$ — half

$4^2 = 16$ — square

$x^2 + 8x + 16 = (x + 4)^2$ — check

Answer: c = 16

2 Solve by completing the square

1. Move the constant: $x^2 + bx = \ldots$

انقل الثابت.

2. Add $(\frac{b}{2})^2$ to both sides.

زوّد للطرفين.

3. Factor, then use the Square Root Property.

حلّل ثم الجذر.

Solve $x^2 - 10x + 14 = 5$.

$x^2 - 10x = -9$ — subtract 14

$x^2 - 10x + 25 = -9 + 25$ — add $(\frac{-10}{2})^2 = 25$

$(x - 5)^2 = 16 \rightarrow x - 5 = \pm 4$ — square root

$x = 9$ **or** $x = 1$

Answer: x = 1 or x = 9

3 When a ≠ 1 · no real solutions

1. Divide every term by **a**.

اقسم على a.

2. Complete the square as usual.

كمّل المربع.

3. If a square equals a negative number → no real solutions.

مربع = سالب ← لا حل.

Solve $-4x^2 + 32x - 72 = 0$.

$x^2 - 8x + 18 = 0$ — divide by −4

$x^2 - 8x + 16 = -18 + 16$ — add 16

$(x - 4)^2 = -2$ — a square cannot be negative

Answer: no real solutions

DON'T FALL FOR IT

✗	✓	
$x^2 - 10x = -9 \rightarrow x^2 - 10x + 25 = -9$	$= -9 + 25$	Add the same number to BOTH sides. زوّد للطرفين.
c for $x^2 + 8x$ is 8^2	$(\frac{8}{2})^2 = 16$	Half of **b** first, then square. النص الأول.

Make It Square

Solving Quadratic Equations by Completing the Square

SCAN FOR PRACTICE

MUST KNOW

4 Write in vertex form (a = 1)

1. Move the constant away from the **x**-terms.

افصل الثابت.

2. Add $(\frac{b}{2})^2$ to both sides.

زوّد للطرفين.

3. Factor and solve for **y**.

حلّل واعزل y.

Write $y = x^2 + 2x - 5$ in vertex form. Find the axis, extremum and zeros.

$y + 5 + 1 = x^2 + 2x + 1$

add 5, then add $(\frac{2}{2})^2 = 1$

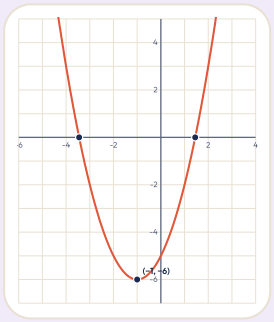
$y = (x + 1)^2 - 6$

vertex (−1, −6), min −6, axis x = −1

$0 = (x + 1)^2 - 6 \rightarrow$
$x = -1 \pm \sqrt{6}$

zeros ≈ 1.45, −3.45

Answer: $y = (x + 1)^2 - 6$



5 Write in vertex form (a ≠ 1)

1. Factor **a** out of the **x**-terms.

خد a برا.

2. Complete the square inside. Remember: you really added $a \cdot (\frac{b}{2})^2$.

اللي اتضاف فعلًا = a × العدد.

3. Balance and simplify.

وازن.

Write $y = -2x^2 + 20x - 42$ in vertex form.

$y + 42 = -2(x^2 - 10x)$

factor −2

$y + 42 - 50 = -2(x^2 - 10x + 25)$

adding 25 inside adds −50

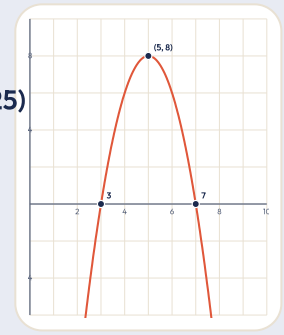
$y = -2(x - 5)^2 + 8$

vertex (5, 8), max 8

$0 = -2(x - 5)^2 + 8 \rightarrow x = 3$ or 7

zeros

Answer: $y = -2(x - 5)^2 + 8$



6 Max or min from vertex form

1. **a > 0**: minimum. **a < 0**: maximum.

إشارة a.

2. The value is **k**. It happens at **x = h**.

القيمة k عند x = h.

A golf ball's height is $y = -(x - 3)^2 + 16$. What is the greatest height and where?

$a = -1 < 0 \rightarrow$ **maximum** — opens down

vertex (3, 16) — read

Answer: 16 at x = 3

DON'T FALL FOR IT

✗ $-2(x^2 - 10x + 25) \rightarrow$ subtract 25	✓ balance with 50	Inside a bracket multiplied by −2, adding 25 really adds −50. اللي اتضاف = −2 × 25.
✗ $y = (x + 1)^2 - 6 \rightarrow$ vertex (1, −6)	✓ (−1, −6)	$(x + 1) = (x - (-1))$, so **h = −1**. h بعكس الإشارة.

LAST LOOK · LESSON 12-5 · THE NIGHT BEFORE THE EXAM

Make It Square

Solving Quadratic Equations by Completing the Square

SCAN FOR PRACTICE

Complete the square

1 Find c

For $x^2 + bx$: $c = (\frac{b}{2})^2$.

$x^2 + 8x + 16 = (x + 4)^2$

نص b ورتّعه.

2 Solve by completing

Move **c**, add $(\frac{b}{2})^2$ to BOTH sides, factor, square root.

$x^2 - 10x = -9 \rightarrow (x - 5)^2 = 16$

زوّد للطرفين.

3 a ≠ 1 · no real roots

Divide by **a** first. If you get $(\ldots)^2$ = negative, there are no real solutions.

$(x - 4)^2 = -2 \rightarrow \varnothing$

اقسم على a. مربع = سالب ← لا حل.

4 Vertex form

Group the **x**-terms, add $(\frac{b}{2})^2$ and subtract it again (or add it to both sides).

$x^2 + 2x - 5 = (x + 1)^2 - 6$

أكمل المربع.

5 Vertex form, a ≠ 1

Factor **a** out of the **x**-terms first. The number you add inside is multiplied by **a**.

$-2(x^2 - 10x + 25)$ adds -50

خد a برا الأول.

6 Read the vertex

Vertex **(h, k)**, axis **x = h**, **a > 0** min, **a < 0** max.

$-2(x - 5)^2 + 8 \rightarrow$ max 8

اقرأ الرأس والمحور.

7 Zeros

Set **y = 0**, then use the Square Root Property.

$(x + 1)^2 = 6 \rightarrow -1 \pm \sqrt{6}$

الأصفار بالجذر.

WORDS TO KNOW

completing the square إكمال المربع	**±** موجب أو سالب	**vertex form** الصورة الرأسية	**extremum (extrema)** قيمة قصوى
zeros الأصفار			

In the final exam: **Q8 · Q11** · scan the QR for timed practice

One Formula Rules

Solving Quadratic Equations by Using the Quadratic Formula

SCAN FOR PRACTICE

MUST KNOW

1 a = 1

1. Standard form, then **a**, **b**, **c**.
حدد المعاملات.
2. Substitute into the formula.
عوّض.
3. Simplify and split into two answers.
بسّط وافصل.

Solve $x^2 - 12x = -11$.

$x^2 - 12x + 11 = 0$: **a = 1, b = −12, c = 11** — standard form

$$x = \frac{12 \pm \sqrt{144 - 44}}{2} = \frac{12 \pm 10}{2}$$ — substitute

x = 11 or x = 1 — split

Answer: x = 1 or x = 11

2 a ≠ 1

1. Do not forget **2a** in the denominator.
2a في المقام.
2. Watch the signs inside $b^2 - 4ac$ when **c** is negative.
انتبه للإشارات.

Solve $2x^2 + 5x = 12$.

$2x^2 + 5x - 12 = 0$: **a = 2, b = 5, c = −12** — standard form

$$x = \frac{-5 \pm \sqrt{25 + 96}}{4} = \frac{-5 \pm 11}{4}$$ — substitute

$$x = \frac{6}{4} = \frac{3}{2} \text{ or } x = \frac{-16}{4} = -4$$ — split

Answer: $x = -4$ or $x = \frac{3}{2}$

3 Real-life (round)

1. Write the equation and move everything to one side.
اكتب المعادلة.
2. Use the formula with a calculator.
الآلة الحاسبة.
3. Keep the answer that makes sense.
اختار المنطقي.

Blood pressure $P = 0.01t^2 + 0.5t + 107$. At what age t is P = 120?

$0.01t^2 + 0.5t - 13 = 0$ — subtract 120

$$t = \frac{-0.5 \pm \sqrt{0.25 + 0.52}}{0.02} = \frac{-0.5 \pm \sqrt{0.77}}{0.02}$$ — formula

t ≈ 18.9 or t ≈ −68.9 — age cannot be negative

Answer: about 18.9 years old

DON'T FALL FOR IT

✗	✓	
$x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a}$	divide EVERYTHING by **2a**	The fraction bar is under **−b** too. الكسر تحت −b كمان.
b = 12 in $x^2 - 12x + 11$	**b = −12**, so **−b = 12**	Take **b** with its sign. b بإشارتها.
$2x^2 + 5x = 12 \rightarrow c = 12$	**c = −12**	Write it **= 0** first. خلّيها = صفر.

One Formula Rules

Solving Quadratic Equations by Using the Quadratic Formula

SCAN FOR PRACTICE

MUST KNOW

4 Number and type of roots

1. Write in standard form and read **a**, **b**, **c**.
حدد المعاملات.
2. Work out $b^2 - 4ac$.
احسب المميز.
3. Use the table.
استخدم الجدول.

Describe the roots of $x^2 - 5x + 2 = 0$.

$(-5)^2 - 4(1)(2) = 25 - 8 = 17$ — discriminant

$17 > 0$ and not a perfect square

Answer: two real irrational roots

5 Exact irrational roots

1. Use the formula.
القانون العام.
2. Simplify the square root: $\sqrt{24} = \sqrt{4 \cdot 6} = 2\sqrt{6}$.
بسّط الجذر.
3. Divide every term by the denominator if possible.
اقسم الكل.

Solve $2x^2 - 4x - 1 = 0$. Give exact answers.

$x = \frac{4 \pm \sqrt{16 + 8}}{4} = \frac{4 \pm \sqrt{24}}{4}$ — formula

$= \frac{4 \pm 2\sqrt{6}}{4} = \frac{2 \pm \sqrt{6}}{2}$ — simplify

≈ 2.22 or -0.22 — decimals

Answer: $x = \frac{2 \pm \sqrt{6}}{2}$

DON'T FALL FOR IT

✗ Discriminant 17 → rational roots	✓ irrational	17 is not a perfect square, so $\sqrt{17}$ is irrational. 17 مش مربع كامل.
✗ $b^2 - 4ac$ for $x^2 + 6x + 9$: $36 + 36 = 72$	✓ $36 - 36 = 0$	$-4ac = -4(1)(9) = -36$. انتبه لإشارة $-4ac$.
✗ $\frac{6 \pm 2\sqrt{5}}{2} = 6 \pm \sqrt{5}$	✓ $3 \pm \sqrt{5}$	Divide BOTH terms by 2. اقسم الحدين.

LAST LOOK · LESSON 12-6 · THE NIGHT BEFORE THE EXAM

One Formula Rules

Solving Quadratic Equations by Using the Quadratic Formula

SCAN FOR PRACTICE

1 Standard form

Write $ax^2 + bx + c = 0$ and read **a**, **b**, **c** WITH their signs.

$x^2 - 12x = -11 \rightarrow x^2 - 12x + 11 = 0$

الصيغة القياسية وحدد a, b, c.

2 The formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

works for every quadratic

القانون العام.

3 Simplify

Work out $b^2 - 4ac$ first, then the square root, then split **+** and **−**.

$\frac{12 \pm 10}{2} \rightarrow 11$ and 1

المميز الأول ثم افصل.

4 Real-life

Round as asked and keep sensible answers.

age ≈ 18.9

قرّب واختار المنطقي.

Quadratic Formula

5 Positive, perfect square

Two real RATIONAL roots.

$x^2 - 2x - 8$: 36

موجب ومربع كامل: جذران نسبيان.

6 Positive, not a square

Two real IRRATIONAL roots.

$x^2 - 5x + 2$: 17

موجب وليس مربع: جذران غير نسبيين.

7 Zero

One real rational (double) root.

$x^2 + 6x + 9$: 0

صفر: جذر واحد.

8 Negative

No real roots.

$2x^2 + 3x + 5$: −31

سالب: لا جذور حقيقية.

WORDS TO KNOW

Quadratic Formula القانون العام	**discriminant** المميز	**rational root** جذر نسبي	**irrational root** جذر غير نسبي
simplest radical form أبسط صورة جذرية			

In the final exam: **Q12 · Q13** · scan the QR for timed practice

UNIT 2 · LESSON 12-7 · BOOK P.739

Where They Meet

Solving Systems of Linear and Quadratic Equations

SCAN FOR PRACTICE

MUST KNOW

1 Solve by graphing

1. Graph the parabola.

ارسم القطع.

2. Graph the line.

ارسم الخط.

3. Read the intersection points and check them.

اقرأ نقاط التقاطع وتحقق.

Solve $y = x^2 + 4x - 1$ and $y = 2x + 2$ by graphing.

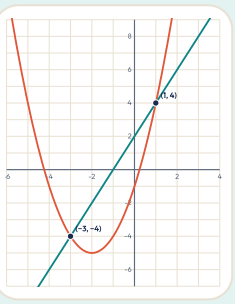
The graphs cross at (−3, −4) and (1, 4)

read

Check (1, 4): $1 + 4 - 1 = 4$ ✓ and $2 + 2 = 4$ ✓

check

Answer: (−3, −4) and (1, 4)



2 Solve algebraically

1. Solve the linear equation for **y**.

حل الخطية ل y.

2. Set the expressions equal and make one side 0.

ساوي وخلّي طرف صفر.

3. Factor (or formula), then find each **y**.

حلّل ثم أوجد y.

Solve $y = x^2 - 2x - 3$ and $x + y = 3$.

$y = -x + 3$ — linear for y

$x^2 - 2x - 3 = -x + 3 \rightarrow x^2 - x - 6 = 0$ — set equal

$(x - 3)(x + 2) = 0 \rightarrow x = 3$ or $x = -2$ — factor

$y = 0$ and $y = 5$ — substitute in $y = -x + 3$

Answer: (3, 0) and (−2, 5)

3 Number of solutions

1. Set the equations equal and write $ax^2 + bx + c = 0$.

ساوي.

2. Find the discriminant: positive → 2, zero → 1, negative → 0.

المميز.

How many solutions: $y = x^2 + 1$ and $y = x - 2$?

$x^2 - x + 3 = 0$ — set equal

$1 - 12 = -11 < 0$ — discriminant

Answer: no solution

4 Real-life systems

1. The question "when are they equal" means set the two models equal.

"متى يتساويان": ساوي.

2. Solve and keep the sensible answer. Read what **x** means (months after January, seconds...).

اختار المنطقي وافهم معنى x.

Swimsuit sales $y = -x^2 + 12x + 25$, gift card sales $y = 5x + 7$ (x = months after January). When are they equal?

$-x^2 + 12x + 25 = 5x + 7 \rightarrow$
$x^2 - 7x - 18 = 0$

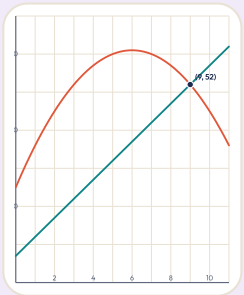
set equal

$(x - 9)(x + 2) = 0 \rightarrow x = 9$ — reject −2

9 months after January

January is x = 0

Answer: October



DON'T FALL FOR IT

✗ Solutions: $x = 3$ and $x = -2$	✓ **(3, 0)** and **(−2, 5)**	A system answer is an ordered pair. Find **y** too. الحل زوج مرتب.
✗ January = month 1, so $x = 9$ → September	✓ October	Read the model: **x** = months PAST January, so January is 0. يناير = صفر.
✗ A line and a parabola always meet twice	✓ 0, 1 or 2 times	Check the discriminant of the combined equation. ممكن صفر أو مرة أو مرتين.

LAST LOOK · LESSON 12-7 · THE NIGHT BEFORE THE EXAM

Where They Meet

Solving Systems of Linear and Quadratic Equations

SCAN FOR PRACTICE

Line + parabola

1 By graphing

Graph both. The solutions are the intersection points.

(−3, −4) and (1, 4)

نقاط التقاطع.

2 Algebraically

Solve both for **y**, set them equal, move to one side, factor or use the formula.

$x^2 + 4x - 1 = 2x + 2$

ساوي الاتنين.

3 Find y

Put each **x** back into the LINEAR equation (easier).

y = 2(1) + 2 = 4

عوّض في المعادلة الخطية.

4 0, 1 or 2 solutions

A line can miss the parabola, touch it once, or cross it twice.

discriminant of the new equation

صفر أو واحد أو اتنين.

WORDS TO KNOW

system of equations نظام معادلات	**intersection** نقطة تقاطع	**substitution** التعويض	**ordered pair** زوج مرتب

In the final exam: **Q14 · Q24** · scan the QR for timed practice

Power Up

Multiplication Properties of Exponents

SCAN FOR PRACTICE

MUST KNOW

1 Is it a monomial?

1. One term only (no + or – between parts).
حد واحد.
2. No variable in a denominator and no negative exponents.
لا متغير في المقام.

Which are monomials: 17, $-4b^2c$, $\frac{ab}{c}$, $x + 2$?

17 ✓, $-4b^2c$ ✓ — one term

$\frac{ab}{c}$ × (divides by a variable), $x + 2$ × (two terms)

Answer: 17 and $-4b^2c$

2 Multiply monomials

1. Group the numbers and each variable.
جمّع الأرقام وكل متغير.
2. Multiply the numbers.
اضرب الأرقام.
3. Add the exponents of the same base.
اجمع الأسس.

Simplify $(7xy^2)(2x^4y^3)$.

$(7 \cdot 2)(x \cdot x^4)(y^2 \cdot y^3)$ — group

$14x^{1+4}y^{2+3} = 14x^5y^5$ — add exponents

Answer: $14x^5y^5$

3 Scientific notation

1. Multiply the decimal parts.
اضرب الأعداد.
2. Add the powers of 10.
اجمع أسس العشرة.
3. Make the first number between 1 and 10.
خلّي العدد بين 1 و10.

A star travels 2,700,000 mph. How far does it go in a year (8760 h)?

$(2.7 \cdot 10^6)(8.76 \cdot 10^3)$ — scientific notation

$= 23.652 \cdot 10^9 = 2.3652 \cdot 10^{10}$ — multiply, adjust

Answer: about $2.4 \cdot 10^{10}$ miles

DON'T FALL FOR IT

✗	✓	
$n^4 \cdot n^7 = n^{28}$	n^{11}	Multiplying powers: ADD the exponents. في الضرب نجمع الأسس.
$(3n^4)(4n^7) = 7n^{11}$	$12n^{11}$	MULTIPLY the coefficients. المعاملات تُضرب.
$x^2 \cdot y^3 = (xy)^5$	x^2y^3	Different bases: you cannot add the exponents. أساسات مختلفة.

Power Up

Multiplication Properties of Exponents

SCAN FOR PRACTICE

MUST KNOW

4 Power of a power

1. Keep the base.

الأساس ثابت.

2. Multiply the exponents.

اضرب الأسس.

Simplify $(x^4)^2$ and $(n^5)^3$.

$(x^4)^2 = x^{4 \cdot 2} = x^8$ — multiply

$(n^5)^3 = n^{15}$ — multiply

Answer: x^8 and n^{15}

5 Power of a product

1. Raise EVERY factor to the power, including the number.

كل العوامل حتى الرقم.

2. Multiply the exponents of each variable.

اضرب الأسس.

3. Watch the sign of a negative number.

انتبه للإشارة.

Simplify $(3x^5y^2)^5$.

$3^5 (x^5)^5 (y^2)^5$ — every factor

$243x^{25}y^{10}$ — simplify

Answer: $243x^{25}y^{10}$

6 Area and volume

1. Write the formula with the side.

اكتب القانون.

2. Use the power of a product.

استخدم القاعدة.

A canvas is a square with side 5x inches. Find its area.

$A = (5x)^2$ — area of a square

$= 5^2 x^2 = 25x^2$ — power of a product

Answer: $25x^2$ in²

DON'T FALL FOR IT

✗	✓	
$(x^4)^2 = x^6$	x^8	Power of a power: MULTIPLY. قوة القوة: نضرب.
$(3x^5)^2 = 3x^{10}$	$9x^{10}$	The number is raised to the power too. الرقم كمان يتربع.
$(-5ab^4)^2 = -25a^2b^8$	$25a^2b^8$	A negative number to an even power is positive. سالب أس زوجي = موجب.

LAST LOOK · LESSON 10-1 · THE NIGHT BEFORE THE EXAM

Power Up

Multiplication Properties of Exponents

SCAN FOR PRACTICE

Multiply powers

1 Monomial?

A number, a variable, or their product, with whole-number exponents. No + or −, no variable in a denominator.

✓ $-5x^2y \times \frac{3}{x}$

حد واحد وأسس كلية.

2 Same base: add

$a^m \cdot a^p = a^{m+p}$.

$n^4 \cdot n^7 = n^{11}$

نفس الأساس: نجمع الأسس.

3 Monomial × monomial

Multiply the numbers, then add the exponents of each letter.

$(3n^4)(4n^7) = 12n^{11}$

اضرب الأرقام واجمع الأسس.

4 Scientific notation

$(2.7 \cdot 10^6)(8.76 \cdot 10^3) = 23.652 \cdot 10^9 = 2.3652 \cdot 10^{10}$.

number between 1 and 10

الصيغة العلمية.

5 Power of a Power

$(a^m)^p = a^{mp}$: MULTIPLY the exponents.

$(n^5)^3 = n^{15}$

نضرب الأسس.

6 Power of a Product

$(ab)^m = a^m b^m$: every factor gets the power, the number too.

$(-5ab^4)^2 = 25a^2b^8$

كل عامل يترفع حتى الرقم.

7 Negative inside

Even power → positive. Odd power → negative.

$(-3x)^3 = -27x^3$

زوجي موجب، فردي سالب.

8 Area · volume

Square: s^2. Cube: s^3. Use the power rules.

$(5x)^2 = 25x^2$

المساحة والحجم.

WORDS TO KNOW

monomial وحيدة الحد	**base · exponent** الأساس · الأس	**power** قوة	**scientific notation** الصيغة العلمية
power of a power قوة القوة	**power of a product** قوة حاصل الضرب	**coefficient** المعامل	

In the final exam: **Q15** · scan the QR for timed practice

UNIT 3 · LESSON 10-2 · BOOK P.579

Subtract the Powers

Division Properties of Exponents

SCAN FOR PRACTICE

MUST KNOW

1 Divide powers and monomials

1. Divide the numbers.
اقسم الأرقام.
2. For each letter: top exponent – bottom exponent.
الأس فوق – الأس تحت.

Simplify $\frac{b^5c^7}{b^2c}$.

$b^{5-2}\,c^{7-1}$ — subtract

b^3c^6

Answer: b^3c^6

2 Power of a quotient

1. Give the power to the numerator and the denominator.
الأس للبسط والمقام.
2. Use the power of a product on each.
ثم قوة حاصل الضرب.

Simplify $(\frac{5p^2}{6})^3$.

$\frac{(5p^2)^3}{6^3}$ — power of a quotient

$\frac{125p^6}{216}$ — simplify

Answer: $\frac{125p^6}{216}$

3 Real-life division

1. "How many times as many" → divide the two numbers.
"كم مرة" = قسمة.
2. Subtract the powers of 10.
اطرح أسس العشرة.

Air at sea level has about 10^{25} molecules per litre, in the stratosphere about 10^{23}. How many times as many at sea level?

$\frac{10^{25}}{10^{23}} = 10^{25-23}$ — divide

$= 10^2 = 100$

Answer: about 100 times

DON'T FALL FOR IT

✗ $\frac{b^6}{b^2} = b^3$	✓ b^4	Quotient of powers: SUBTRACT, do not divide. نطرح الأسس مش نقسمها.
✗ $(\frac{5p^2}{6})^3 = \frac{125p^6}{6}$	✓ $\frac{125p^6}{216}$	The denominator also gets the power. المقام كمان.
✗ $\frac{10^{25}}{10^{23}} = 1^2$	✓ 10^2	The base stays 10. الأساس ثابت.

LAST LOOK · LESSON 10-2 · THE NIGHT BEFORE THE EXAM

Subtract the Powers

Division Properties of Exponents

SCAN FOR PRACTICE

1 Quotient of Powers

$\frac{a^m}{a^p} = a^{m-p}$: SUBTRACT the exponents.

$\frac{b^5}{b^2} = b^3$

نطرح الأسس.

2 Divide monomials

Divide the numbers, subtract the exponents of each letter.

$\frac{a^7b^5c^4}{a^2b^3c} = a^5b^2c^3$

اقسم الأرقام واطرح الأسس.

Quotients of powers

3 Power of a Quotient

$(\frac{a}{b})^m = \frac{a^m}{b^m}$.

$(\frac{5p^2}{6})^3 = \frac{125p^6}{216}$

البسط والمقام لهم الأس.

4 Real-life

$\frac{10^{25}}{10^{23}} = 10^2 = 100$ times as many.

"how many times" → divide

"كم مرة" = قسمة.

WORDS TO KNOW

quotient	ناتج القسمة
Quotient of Powers	قسمة القوى
Power of a Quotient	قوة ناتج القسمة

In the final exam: **Q16** · scan the QR for timed practice

UNIT 3 · LESSON 10-3 · BOOK P.587

Flip It Over

Negative Exponents

SCAN FOR PRACTICE

MUST KNOW

1 Zero exponent

1. Anything (not 0) to the power 0 is 1.

أي شيء أس صفر = 1.

2. Only the part under the 0 becomes 1.

بس الجزء اللي عليه الصفر.

Simplify $\frac{a^4b^0}{a^2}$.

$b^0 = 1$ — zero exponent

$\frac{a^4}{a^2} = a^2$ — quotient of powers

Answer: a^2

2 Negative exponents

1. A factor with a negative exponent moves to the other side of the fraction bar.

يتنقل للناحية التانية.

2. Its exponent becomes positive.

والأس يبقى موجب.

3. Numbers without a negative exponent stay where they are.

الباقي مكانه.

Simplify $\frac{a^7b^{-4}}{c^{-2}}$.

$b^{-4} = \frac{1}{b^4} \cdot \frac{1}{c^{-2}} = c^2$ — move

$\frac{a^7c^2}{b^4}$ — positive exponents

Answer: $\frac{a^7c^2}{b^4}$

3 Mixed simplifying

1. Group the numbers and each base.

جمّع.

2. Use the quotient/product rules.

القواعد.

3. Finish with only positive exponents.

أسس موجبة في الآخر.

Simplify $\frac{3g^{-3}h^2}{-36g^3h^{-4}}$.

$\frac{3}{-36} = -\frac{1}{12}$ — numbers

$g^{-3-3} = g^{-6} \cdot h^{2-(-4)} = h^6$ — exponents

$-\frac{h^6}{12g^6}$ — positive exponents

Answer: $-\frac{h^6}{12g^6}$

DON'T FALL FOR IT

× $3^{-2} = -9$	✓ $\frac{1}{9}$	A negative exponent means reciprocal, not a negative number. الأس السالب مقلوب مش سالب.
× $5x^0 = 1$	✓ 5	Only **x** has the exponent 0. الصفر على x بس.
× $2x^{-3} = \frac{1}{2x^3}$	✓ $\frac{2}{x^3}$	The 2 has no negative exponent, so it stays on top. 2 مكانها.

LAST LOOK · LESSON 10-3 · THE NIGHT BEFORE THE EXAM

Flip It Over

Negative Exponents

SCAN FOR PRACTICE

Zero & negative exponents

1 Zero exponent

$a^0 = 1$ for any $a \neq 0$.

$\left(\frac{-8m^2np^5}{9k^2mn^4}\right)^0 = 1$

أي عدد أس صفر = 1.

2 Negative exponent

$a^{-n} = \frac{1}{a^n}$ and $\frac{1}{a^{-n}} = a^n$. Move it across the fraction bar and make the exponent positive.

$3^{-5} = \frac{1}{243}$

الأس السالب يقلب مكانه.

3 Simplified means

Only positive exponents, each base once, no powers of powers, fractions in simplest form.

$\frac{a^7c^2}{b^4}$

أسس موجبة وكل أساس مرة.

4 Order of magnitude

Round to the nearest power of 10 and compare.

$\frac{10^{-2}}{10^2} = 10^{-4}$

المرتبة العشرية.

WORDS TO KNOW

zero exponent	الأس الصفري
negative exponent	الأس السالب
reciprocal	المقلوب
order of magnitude	المرتبة العشرية

In the final exam: Q17 · scan the QR for timed practice